

The Gradient of the Logistic Function

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1 Introduction: Logistic Regression

2 Loss

To compute the loss function, we want to determine each individual probabilities distance from the true result. For a single observation i it takes the form.

$$(1) \quad \text{Loss}_i = - \left[\frac{y_i}{\log(p_i)} + \frac{1 - y_i}{\log(1 - p_i)} \right]$$

In general for n observations we can think of y and p as vectors each of length n , where if y takes on the value 1 then the left side is realized and if y takes on the value 0 then the right side is realized ie $1 - 1 = 0$ then the right side is $0 / \log(1 - p) = 0$.

We calculate a average loss in the form.

$$(2) \quad \text{Average Loss} = \frac{1}{n} \sum_{i=1}^n \text{Loss}_i$$

Or in other words take the average over the n observations.

3 Gradient

The gradient is simply a vector of values in which we would like to move each w_j in order to minimize error the fastest. It can be thought of geometrically as

the direction of steepest slope. So we want to step in that direction to move as efficiently as possible.

We want the change in loss for a change in w_j meaning we want the partial derivative in the loss in terms of the change in each weight. Specifically we want $\frac{\partial L_i}{\partial w_{ij}}$